

# Nano-Tek: First Firmware Flash

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Builder's guide for all board revisions · October 2026

A freshly built Nano-Tek has a blank STM32. You flash FlashFloppy onto it **once**, with a USB-serial adapter or an ST-Link. After that, FlashFloppy updates itself from a USB stick.

**USB-C can't do the first flash.** The STM32's USB programming mode (DFU) does not start on a blank chip. Use one of the options below.

Steps that differ between boards are marked **Only applies to Rev 1.0-1.2** or **Only applies to Rev 1.3 and later**. The revision is printed on the back of the board.

## What you need

**Firmware.** From the [FlashFloppy releases page](#), download the release file `flashfloppy-<version>.zip`, not "Source code (zip)". Inside it you need:

- `hex/flashfloppy-at415-st105-<version>.hex` for the first flash.
- `flashfloppy-<version>.upd` for later updates from a USB stick.

**One programmer** (any one of these):

| Option      | Programmer                                                | Connects to | Software                                                    |
|-------------|-----------------------------------------------------------|-------------|-------------------------------------------------------------|
| A (easiest) | 3.3 V USB-serial adapter: CP2102, CH340, FT232 or FT2232H | J2          | STM32CubeProgrammer (Windows, macOS, Linux) or stm32flash   |
| B           | PL2303 USB-serial cable (4 wires)                         | J2          | Same as A; Windows 10/11 may need the PL2303 Legacy Updater |
| C           | Cheap ST-Link V2 clone (USB stick shape)                  | J4          | STM32 ST-LINK Utility or OpenOCD                            |
| D           | Official ST-LINK/V2 (20-pin ribbon)                       | J4          | STM32CubeProgrammer                                         |

**Also:** 3–4 female DuPont jumper wires, one 2.54 mm jumper cap, and power for the board: a USB-A to USB-C cable from a PC or USB charger, or the floppy power cable. **USB-C to USB-C cables don't power the board.**

**Downloads:** [FlashFloppy releases](#) · [FlashFloppy wiki](#) · [STM32CubeProgrammer](#) · [ST-LINK driver \(STSW-LINK009\)](#) · [STM32 ST-LINK Utility \(STSW-LINK004\)](#) · [OpenOCD](#), [xPack build](#) · [stm32flash](#) · [PL2303 Legacy Updater](#) · [ST-LINK/V2 manual UM1075](#)

## The two headers

**J2 (2×5)** **Only applies to Rev 1.0-1.2**

|           |             |              |            |             |
|-----------|-------------|--------------|------------|-------------|
| 2<br>GND  | 4<br>+5V    | 6<br>GND     | 8<br>GND   | 10<br>+3.3V |
| 1<br>NRST | 3<br>PA9 TX | 5<br>PA10 RX | 7<br>+3.3V | 9<br>BOOT0  |

RX and TX are swapped on these boards and **don't match the silkscreen**. Wire by this diagram, not by the board labels.

On every revision: jumper cap on J2 9–10 = BOOT0 high (programming mode). Pin 4 (+5V) is only for optional bench power (Option B). Pin numbers follow the schematic; pin 1 has the square pad.

**J2 (2×5)** **Only applies to Rev 1.3 and later**

|           |              |             |            |             |
|-----------|--------------|-------------|------------|-------------|
| 2<br>GND  | 4<br>+5V     | 6<br>GND    | 8<br>GND   | 10<br>+3.3V |
| 1<br>NRST | 3<br>PA10 RX | 5<br>PA9 TX | 7<br>+3.3V | 9<br>BOOT0  |

Matches the silkscreen. Labels name the board's own pins, so the adapter's TX goes to the pin marked RX.

#### J4: encoder + SWD header, all revisions

|              |             |             |            |          |            |            |           |
|--------------|-------------|-------------|------------|----------|------------|------------|-----------|
| 1<br>ENC CLK | 2<br>ENC DT | 3<br>ENC SW | 4<br>+3.3V | 5<br>GND | 6<br>SWDIO | 7<br>SWCLK | 8<br>NRST |
|--------------|-------------|-------------|------------|----------|------------|------------|-----------|

Encoder on 1-5, SWD on 6-7; with the encoder plugged in, take GND from J2-2. **Only applies to Rev 1.3 and later** pin 8 = NRST (dashed); Rev 1.0-1.2 have 7 pins.

##### Power rules: read before plugging anything in

- Only applies to Rev 1.0-1.2** Power the board from USB-C **or** from the floppy power connector (CN1), **never both**. USB-C power is wired straight to the board's 5 V, so the PC and the Amiga supplies would feed into each other.
- Only applies to Rev 1.3 and later** USB-C feeds the board through a diode, and the USB power switch stays off while you program, so CN1 may stay connected. Don't plug a PC into USB-C while FlashFloppy is running from CN1 power.
- Never use a programmer's 3.3 V pin as power. A serial cable's 5 V lead may go to J2-4, but only with CN1 and USB-C unplugged.
- BOOT0 is only read at power-up: fit the jumper cap **before** powering the board.

### Option A: USB-serial adapter (recommended)

This uses the bootloader built into every STM32: three wires and the BOOT0 jumper. Set the adapter to **3.3 V** logic (many have a 3.3/5 V jumper).

| Adapter | J2 on Rev 1.0-1.2 | J2 on Rev 1.3 and later |
|---------|-------------------|-------------------------|
| TX      | pin 5 (PA10 RX)   | pin 3 (PA10 RX)         |
| RX      | pin 3 (PA9 TX)    | pin 5 (PA9 TX)          |
| GND     | pin 2             | pin 2                   |

Don't connect the adapter's 3.3 V or 5 V pins. On an FT2232H board, use channel A: AD0 = TX, AD1 = RX.

- Fit the jumper cap on J2 pins 9-10 (BOOT0).
- Power the board. The OLED stays blank, which is expected.
- Open STM32CubeProgrammer and select **UART**. Choose the adapter's COM port, baud rate 115200, parity **Even**, 8 data bits, 1 stop bit, flow control off. Click **Connect**. You should see Chip ID **0x418**.
- Open **Erasing & Programming**, browse to the `.hex` file, tick **Verify programming**, and click **Start Programming**.
- Click **Disconnect**, remove the jumper cap and the wires, and power-cycle. FlashFloppy appears on the OLED.

Command-line versions (replace the port and version):

```
STM32_Programmer_CLI -c port=COM5 br=115200 P=EVEN -w flashfloppy-at415-st105-3.44.hex -v  
stm32flash -b 115200 -w flashfloppy-at415-st105-3.44.hex -v /dev/ttyUSB0
```

Tips: if Connect fails, power-cycle the board before retrying, because the bootloader locks onto the first interface it hears from. If it still fails, swap TX and RX.

## Option B: PL2303 USB-serial cable

The common 4-wire PL2303 cable works just like Option A. On Windows 10/11, older and clone PL2303 chips show **Code 10** or "PL2303HXA PHASED OUT" in Device Manager until you give them a legacy driver:

1. Install the [PL2303 Legacy Updater](#), plug in the cable, run "PL2303 Legacy Updater" and pick the driver it offers (3.3.11.152 for HXA/XA chips, 3.8.36.2 for TA/TB).
2. Device Manager → Ports should now show "Prolific USB-to-Serial Comm Port (COMx)" with no warning. If Windows Update swaps the driver back later, run the updater again.

| Cable wire (usual color) | J2 on Rev 1.0-1.2 | J2 on Rev 1.3 and later |
|--------------------------|-------------------|-------------------------|
| TX (green)               | pin 5             | pin 3                   |
| RX (white)               | pin 3             | pin 5                   |
| GND (black)              | pin 2             | pin 2                   |
| +5V (red)                | pin 4 (optional)  | pin 4 (optional)        |

Wire colors vary between cables, so check yours. The red +5V lead on J2-4 can power the board on its own, which is handy on the bench, but only with CN1 and USB-C unplugged; otherwise leave it unconnected. 3.3 V and 5 V logic cables both work (PA9 and PA10 are 5 V tolerant). Then program exactly as in Option A, steps 1-5.

## Option C: cheap ST-Link V2 clone (SWD)

A blank chip accepts SWD straight away: no BOOT0 jumper and no serial wires needed. Wire by the labels printed on the clone's case.

| Clone | Nano-Tek       |
|-------|----------------|
| GND   | J4-5 (or J2-2) |
| SWDIO | J4-6           |
| SWCLK | J4-7           |

- Leave the clone's 3.3 V and RST pins unconnected. Power the board from USB-C or CN1.
- Install ST's ST-LINK driver (STSW-LINK009).
- **STM32 ST-LINK Utility:** Target → Connect, then Target → Program & Verify. Pick the `.hex`, tick Verify and Reset after programming, then Start.
- **OpenOCD** (xPack build), one command:

```
openocd -f interface/stlink.cfg -f target/stm32f1x.cfg -c "adapter speed 100" -c "program flashflop  
y-at415-st105-3.44.hex verify reset exit"
```

- STM32CubeProgrammer often refuses clones. Don't let any ST tool "upgrade" a clone's firmware. Clones built on non-ST chips can be bricked.
- Only one program can use the ST-Link at a time; close the others first.

## Option D: official ST-LINK/V2 (20-pin)

Same as Option C plus one wire: the probe must see the board's 3.3 V on its pin 1 (VAPP), or it won't connect. It does not power the board.

| ST-LINK/V2 pin           | Signal                      | Nano-Tek                           |
|--------------------------|-----------------------------|------------------------------------|
| 1                        | VAPP (target voltage sense) | J4-4 or J2-7 (+3.3V), required     |
| 4 (or any even pin 4-20) | GND                         | J4-5 or J2-2                       |
| 7                        | SWDIO                       | J4-6                               |
| 9                        | SWCLK                       | J4-7                               |
| 15                       | NRST (optional)             | J2-1, or J4-8 on Rev 1.3 and later |
| 19                       | Probe's own 3.3 V output    | <b>Do not connect</b>              |

Pin 1 is on the red-stripe side of the ribbon cable. In STM32CubeProgrammer choose **ST-LINK**, port SWD, mode Normal, click Connect, then program as in Option A step 4. A printable label for the probe is in the Nano-Tek repository: [docs/ST-LINK-V2\\_Label.pdf](#).

## After the first flash

**Updates.** Copy `flashfloppy-<version>.upd` to the root of the USB stick (only one `.upd` file; two give error E02). Hold the encoder button while powering on and follow the OLED.

**Re-flashing with SWD later.** FlashFloppy switches SWD off when it runs, so fit the BOOT0 jumper before powering the board.

**Rescue over USB-C.** Once FlashFloppy has been flashed, the chip stays non-blank (updates never erase FlashFloppy's updater), so USB-C programming works for recovery:

1. **Only applies to Rev 1.0-1.2** Tie PA9 to +3.3V with a DuPont wire from J2-3 to J2-7, and unplug the floppy power (CN1). Rev 1.3 and later have the PA9 pull-up on the board, so no wire is needed.
2. Fit the jumper cap on J2 9-10. Connect USB-C to the PC with a USB-A to USB-C **data** cable. Charge-only cables show nothing.
3. Windows shows **STM32 BOOTLOADER**. In STM32CubeProgrammer choose **USB**, Connect, and program the `.hex` as above.
4. Remove the jumper cap (and the PA9 wire), then power-cycle.

## Troubleshooting

| Problem                                           | Fix                                                                                                                                                 |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Serial: Connect times out                         | Check TX/RX against the diagram for your board revision, or swap them. Check BOOT0 is fitted. Power-cycle, then Connect again. Parity must be Even. |
| OLED shows FlashFloppy while you try to program   | BOOT0 wasn't fitted at power-up. Fit it and power-cycle.                                                                                            |
| ST-Link: "unable to connect to the target"        | Check the three wires and the clone's labels; try swapping SWDIO/SWCLK. If FlashFloppy is already on the chip, fit BOOT0.                           |
| OpenOCD: "LIBUSB_ERROR_ACCESS"                    | Another program has the ST-Link open. Close STM32CubeProgrammer and ST-LINK Utility.                                                                |
| Official ST-LINK shows 0 V target                 | Pin 1 (VAPP) isn't wired to +3.3V.                                                                                                                  |
| USB-C: nothing appears on the PC                  | Blank chip (use Option A, B, C or D), charge-only cable, missing PA9-to-3.3V wire (Rev 1.0-1.2), USB-C to USB-C cable, or BOOT0 not fitted.         |
| Update stops with E02                             | More than one <code>.upd</code> file on the stick. Keep one.                                                                                        |
| PL2303: Code 10 or "PHASED OUT" in Device Manager | Run the PL2303 Legacy Updater and pick the legacy driver; run it again after Windows updates.                                                       |